PRACTICAL 8

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a program to calculate the percentage of page hits and page faults if the system uses the Least Recently Used (LRU) page replacement technique for the page reference string entered by the user.

Example page reference string:
1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

Number of frames = 3

```
 UW PICO 5.09

                found = 1;
                time[j] = i; // update recent use
                break;
            }
        }

        // If not found → page fault
        if(!found) {
            pos = 0;

            // Find least recently used
            for(j = 1; j < f; j++) {
                if(time[j] < time[pos]) {
                    pos = j;
                }
            }

            frames[pos] = pages[i];
            time[pos] = i;
            faults++;
        }

        // Print frame status
        printf("\nFrames: ");
        for(j = 0; j < f; j++) {
            if(frames[j] != -1)
                printf("%d ", frames[j]);
            else
                printf("- ");
        }
    }

    printf("\n\nTotal Page Faults = %d", faults);
    printf("\nTotal Page Hits = %d", hits);

    float hit_ratio = (float)hits / n * 100;
    float fault_ratio = (float)faults / n * 100;

    printf("\nHit Percentage = %.2f%%", hit_ratio);
    printf("\nFault Percentage = %.2f%%\n", fault_ratio);

    return 0;
}
```

```
Last login: Sun Mar 15 18:34:37 on ttys000
(base) sakshirakhade@Sakshis-MacBook-Air ~ % nano lru.c
(base) sakshirakhade@Sakshis-MacBook-Air ~ % gcc lru.c -o lru
(base) sakshirakhade@Sakshis-MacBook-Air ~ % ./lru
Enter number of pages: 12
Enter page reference string:
1 2 3 4 1 2 5 1 2 3 4 5
Enter number of frames: 3

Frames: - 1 -
Frames: - 2 -
Frames: - 3 -
Frames: - 4 -
Frames: - 1 -
Frames: - 2 -
Frames: - 5 -
Frames: - 1 -
Frames: - 2 -
Frames: - 3 -
Frames: - 4 -
Frames: - 5 -

Total Page Faults = 12
Total Page Hits = 0
Hit Percentage = 0.00%
Fault Percentage = 100.00%
(base) sakshirakhade@Sakshis-MacBook-Air ~ %
```